

Practical 4: Heirarchical work processing

Participants:

Heinrich Muller u25050029

Amelton Khoza u1920162

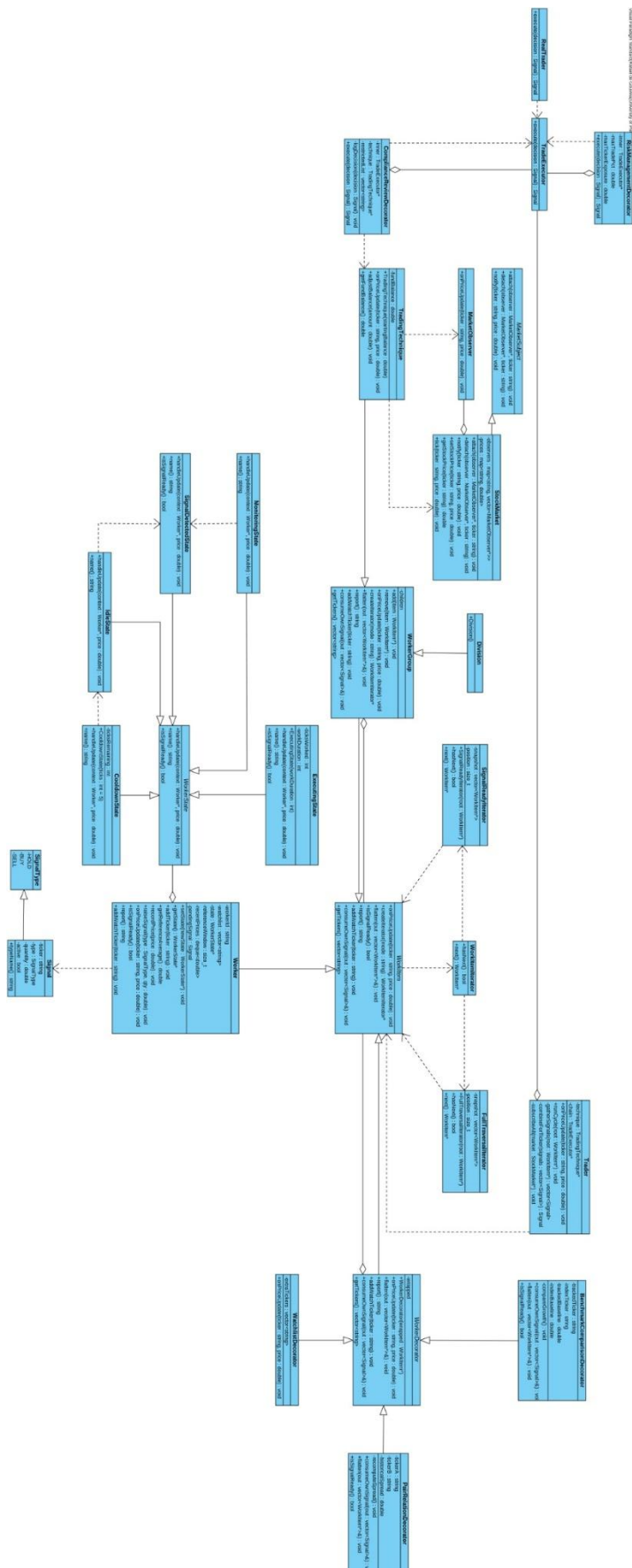
Rafael de Gouvenia u25054903

System Description:

TaskForge is an automated, event-driven trading-strategy platform designed to organize and execute quantitative market operations. In a real-world trading firm, market evaluation, correlation tracking, risk enforcement, compliance auditing, and order routing must operate collaboratively without rigid, hard-coded dependencies.

TaskForge addresses this by structuring the entire organization into a recursive, part-whole hierarchy that ingests live market ticks, routes data safely across departments, tracks the lifecycles of individual analysts, and validates every trade through runtime-configurable decorator pipelines.

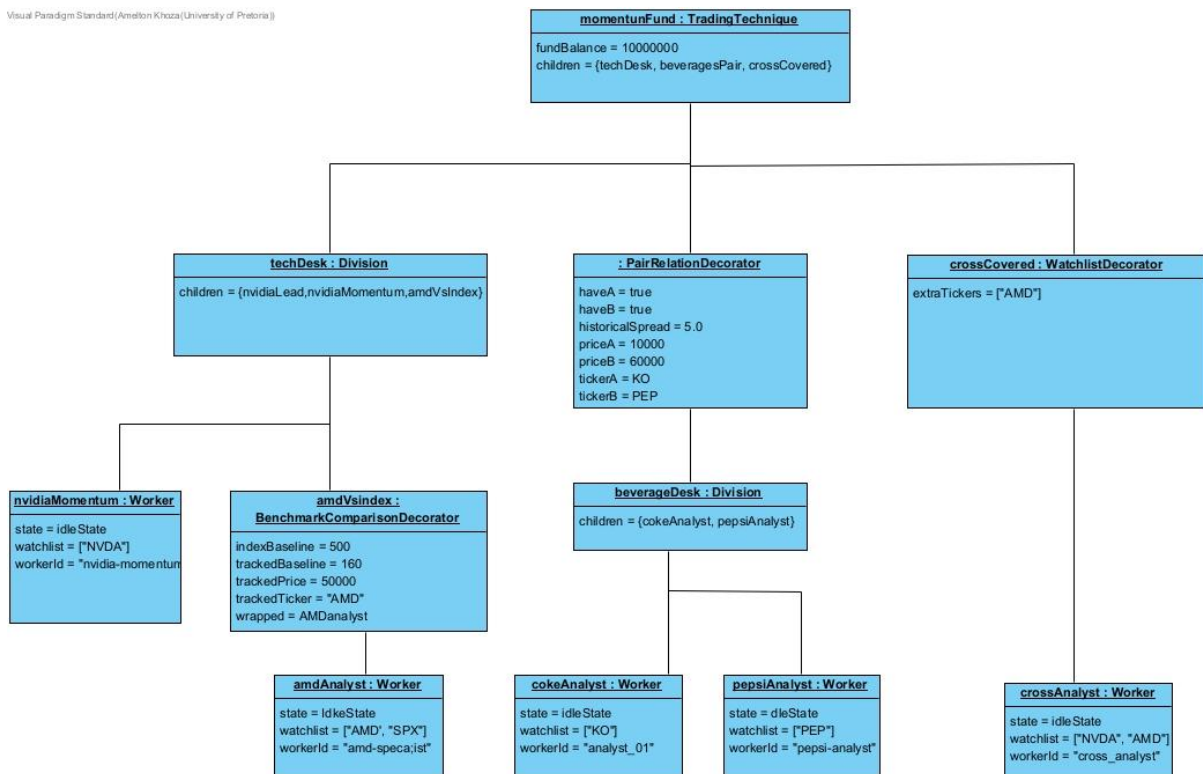
UML (Class Diagram)



All 5 diagrams:

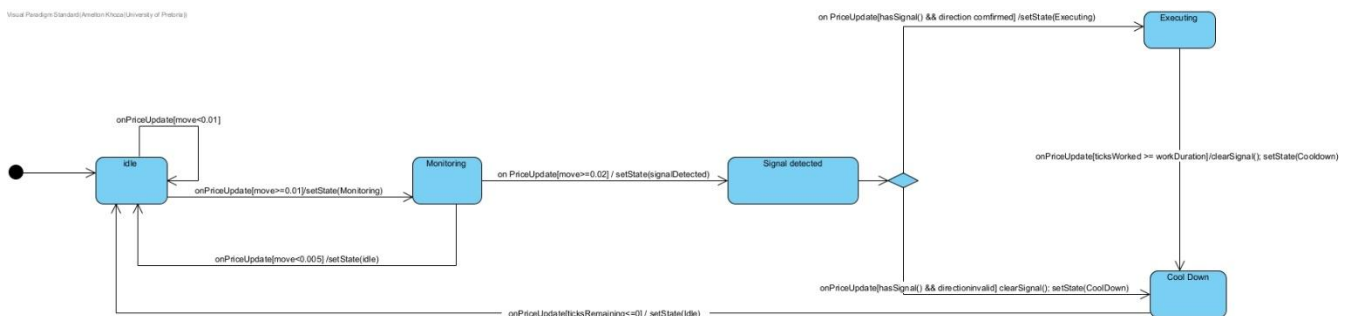
1. Object Diagram:

Visual Paradigm Standard(Amelton Khoza(University of Pretoria))

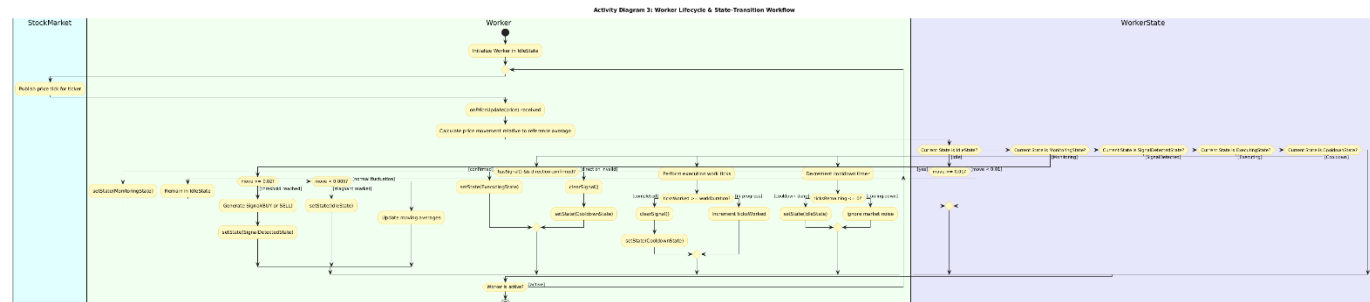


2. State diagram:

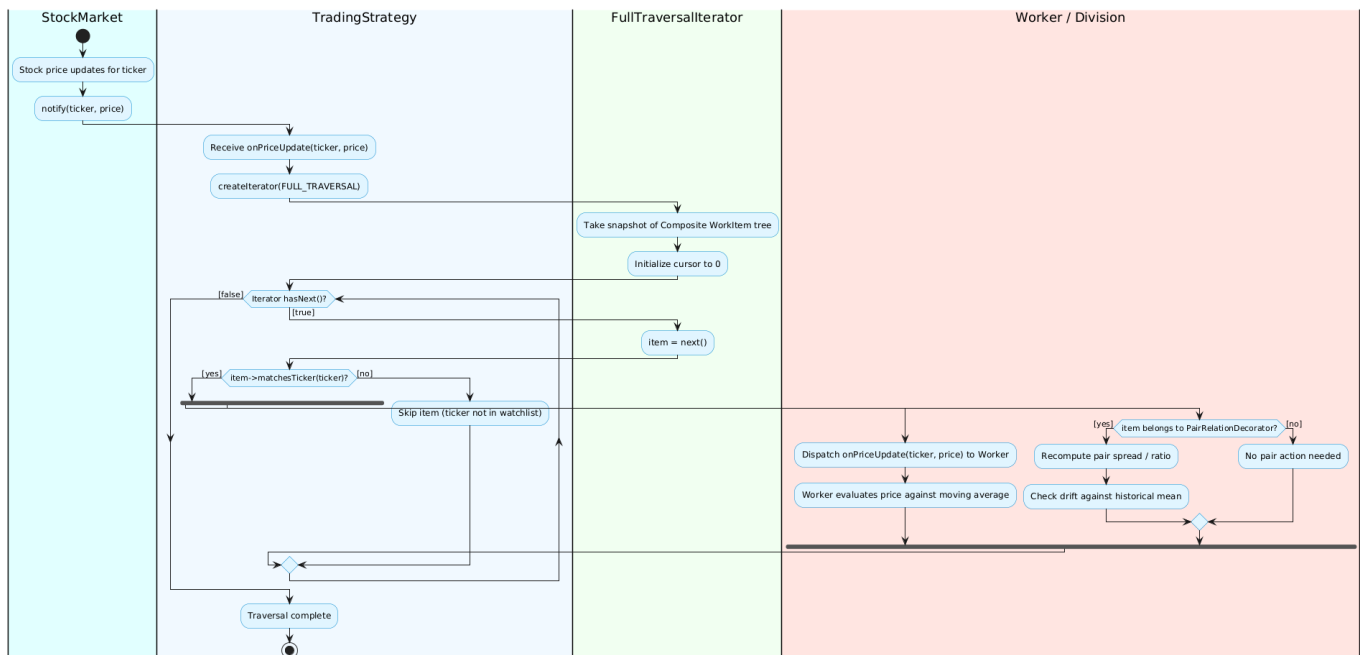
Visual Paradigm Standard(Amelton Khoza(University of Pretoria))



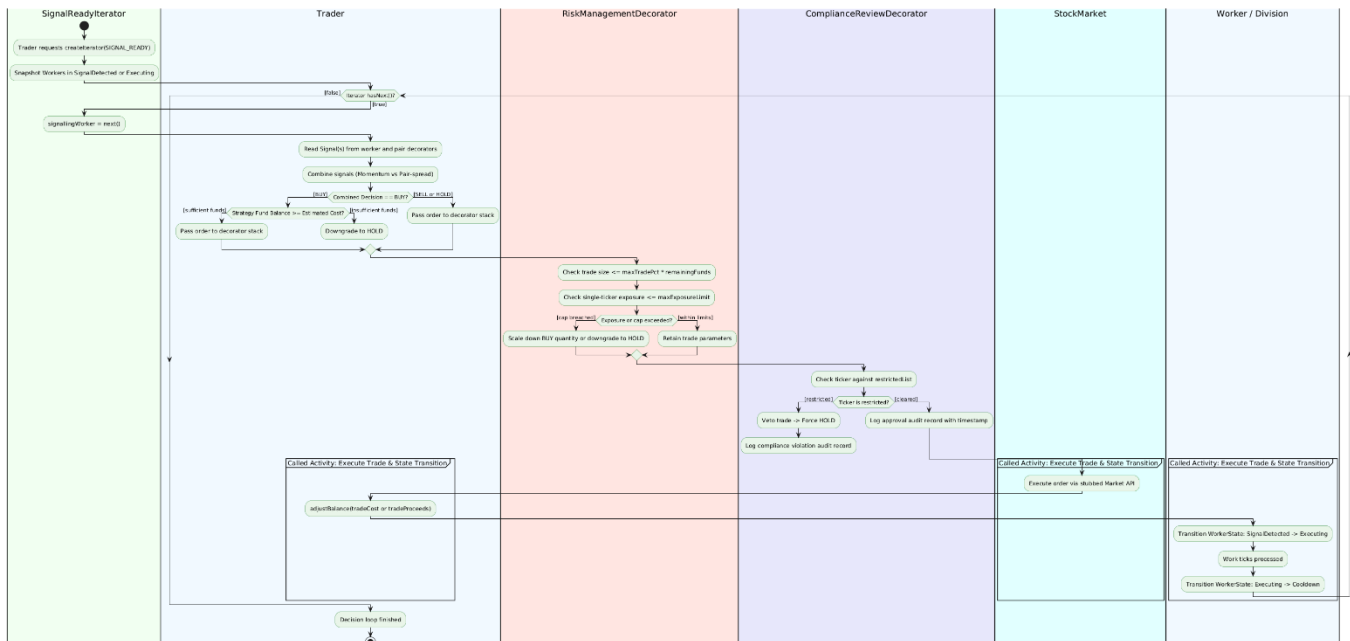
3. Activity Diagram (state):



4. Activity diagram(Iterator):



5. Activity diagram(observer):



Git hub Link

Click on link :[link to github](#)

Ownership rationale

We built a stock market simulation, it was planned out roughly before any real code was written, so the whole class list shows up early as empty stubs. Most of the design work went into how the object tree gets walked and created. There is a tree of work items, with workers grouped into divisions and some of them wrapped in decorators, and it has to be walked twice. Once at startup to find out which tickers to subscribe to, and again on every market tick to gather trading signals. The iterator classes are written to handle this, but at first only one of those two jobs actually uses them. A later commit pulls the iterator out of the ticker lookup and has the tree report its tickers upward instead, this was done for debugging purposes. When that situation was solved this was reversed and taken further, with the collection recursion stripped out of the tree so the iterator becomes the only thing that knows how to walk it. This leads us to the key design, broadcast pushes down through the tree, while collection pulls across a filtered snapshot. The main design idea in the final version is that a decorator can add itself into the walk. The pair-trading and benchmark decorators create signals that do not belong to any single worker, so they put themselves on the list of nodes to visit only when they have a signal to report.

fsdsf

Contribution statement:

Amelton Khoza:

Worked on the Observer pattern implementation which covered, the MarketObserver, MarketSubject along with the TradingTechnique. I worked on 3 Diagrams names The object, state along with the activity state diagram. I compiled the PDF ensuring that all the requirements are included.

Rafael de Gouvenia:

The Composite pattern implementation, covering WorkItem, WorkerGroup, Division, and TradingTechnique. I also built the full UML class diagram in Visual Paradigm, covering all four required patterns and their relationships, and produced the Decorator activity diagram as part of our UML portfolio. I set up the project's Dockerfile so the team can build, run, and debug the application consistently through Docker, including GDB and Valgrind support, and pushed the resulting header/implementation files, the Dockerfile, and the diagram to our GitHub repository, as well as the makefile.

Heinrich Muller:

Implementation of the Worker State System, Trader, Decorators. Redesign of how signals are processed. Removal of un-used out of scope functions, reformat to ensure we follow the designed UML. Spearheading of overall design.

Valgrind report:

valgrind --leak-check=full --show-leak-kinds=all ./taskforge

==2595== Memcheck, a memory error detector

==2595== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.

==2595== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info

==2595== Command: ./taskforge

==2595==

Stock Market Simulation

All tickers (with duplicates): NVDA NVDA AMD SPX KO PEP NVDA AMD

Unique tickers: NVDA AMD SPX KO PEP

Event distribution over 200 ticks:

CALM: 80

STIR: 50

RALLY: 24

SELLOFF: 16

HOLD_HIGH: 20

HOLD_LOW: 10

Starting balance: 1e+06

Running 200 market ticks across 4 instruments...

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midpoint runtime change (tick 100)

added 'late-analyst' (watchlist + benchmark decorated) to the fund

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resuming run

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[Compliance] Logged BUY KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled BUY NVDA qty=1 (pending compliance)
[Compliance] Logged BUY NVDA qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled SELL KO qty=1 (pending compliance)
[Compliance] Logged SELL KO qty=1
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)

[Compliance] Logged SELL AMD qty=1
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1
[Trader] Settled SELL AMD qty=1 (pending compliance)
[Compliance] Logged SELL AMD qty=1
[Trader] Settled BUY KO qty=2 (pending compliance)
[Compliance] Logged BUY KO qty=2
[Trader] Settled BUY KO qty=1 (pending compliance)
[Compliance] Logged BUY KO qty=1

Final balance: 996600

Last SPX price seen by the exchange: 5000

Done

==2595==

==2595== HEAP SUMMARY:

==2595== in use at exit: 0 bytes in 0 blocks

==2595== total heap usage: 2,538 allocs, 2,538 frees, 194,080 bytes allocated

==2595==

==2595== All heap blocks were freed -- no leaks are possible

==2595==

==2595== For lists of detected and suppressed errors, rerun with: -s

==2595== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)